

Fock-Resolved Qubit-State Inference for a Black-Box SQR Gate: What Wait-and-Displacement Tomography Can and Cannot Recover

Autonomous cQED Research Platform

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Abstract

We study whether a black-box selective qubit rotation can be validated without reconstructing the full joint qubit-cavity process. The assumed calibrated analysis operations are deliberately limited to a cavity displacement, free dispersive waiting under known (χ, χ') , and ordinary qubit tomography. The central target is the effective Fock-resolved qubit output state $\rho_q^{(n, \text{out})} \propto \langle n | \rho_{\text{out}} | n \rangle$ on the low-lying cavity manifolds. We combine an analytic identifiability analysis, single-qubit maximum-likelihood tomography validation, synthetic inverse-problem benchmarks, and pulse-level black-box case studies generated with `cqed_sim`. The positive result is that the weighted transverse sector content $u_n = p_n(X_n + iY_n)$ is recoverable: wait-only tomography is already full-rank on that subspace, while the combined displacement-plus-wait protocol is modestly more noise-robust in the tested grids. The negative result is that the allowed measurements do not separate the Fock populations p_n from the sectorwise longitudinal components Z_n ; displacement-only measurements reduce to the ordinary qubit marginal, and an explicit gauge family shows that distinct sets of $\{p_n, \rho_q^{(n)}\}$ generate identical recoverable data. The combined protocol is nevertheless valuable because it turns cavity coherences into a diagnostic residual: in an exact coherent benchmark the wait-only diagonal-model residual remains at 1.6×10^{-16} , whereas the combined residual rises to 1.39×10^{-1} . Pulse-level black-box tests show the same pattern. For near-diagonal pulse-level outputs the combined protocol recovers the weighted transverse sectors with errors below 3×10^{-3} , but for coherent or leaky outputs its large residual warns that the fitted Fock-resolved picture is only an effective diagnostic. The experimentally defensible recommendation is therefore to use the combined protocol when one wants both recovery of weighted transverse sector data and a coherence/mismatch witness, while avoiding any claim of direct p_n or Z_n recovery unless an additional cavity-sensitive measurement primitive is added.

1 Introduction

Dispersive circuit quantum electrodynamics provides a natural route to photon-number-conditioned qubit control, oscillator encodings, and selective diagnostics of joint qubit-cavity dynamics [1, 2]. In practice, however, a laboratory black-box gate may be available only through its calibrated input-output behavior rather than through a trusted microscopic pulse model. The question addressed here is not how to reconstruct the full joint process, but instead how much of the low-lying Fock-resolved qubit action can be inferred from a restricted set of post-gate analysis operations.

The allowed operations are intentionally weak: a calibrated displacement $D(\alpha)$ on the cavity, free waiting under known dispersive parameters (χ, χ') , and calibrated qubit tomography rotations followed by the usual qubit readout. These ingredients are realistic in many cQED settings and fit

comfortably within standard qubit-state tomography practice [3, 4]. They are also weaker than a fully selective Fock-tagging primitive. That weakness is scientifically useful because it forces the inverse problem to reveal exactly what information is and is not available from such black-box validation.

The study therefore asks a sharply constrained question:

Given a black-box SQR, known dispersive waiting, calibrated displacement, and ordinary qubit tomography, can one reliably infer the effective qubit output state associated with each low-lying cavity Fock sector?

The answer is mixed. A recoverable subspace exists and is experimentally useful, but it is smaller than the optimistic target of full sector-resolved qubit-state tomography.

2 Problem Statement and Inference Targets

We consider a post-gate joint state on a qubit plus cavity. Under the favorable Fock-diagonal model,

$$\rho_{\text{out}} = \sum_{n=0}^{N-1} p_n |n\rangle\langle n| \otimes \rho_q^{(n)}, \quad (1)$$

the sector state is parameterized as

$$\rho_q^{(n)} = \frac{1}{2} (I + X_n \sigma_x + Y_n \sigma_y + Z_n \sigma_z). \quad (2)$$

The more general block model,

$$\rho_{\text{out}} = \sum_{m,n=0}^{N-1} |m\rangle\langle n| \otimes A_{mn}, \quad (3)$$

allows cavity coherences through the off-diagonal blocks A_{mn} with $m \neq n$.

The scientific target is not the full joint state in Eq. (3). Instead, we ask whether the measurements can recover the effective sector marginals in Eq. (1). It is therefore essential to distinguish three different inference goals:

1. recovery of the weighted transverse sector content $p_n X_n$ and $p_n Y_n$;
2. recovery of the normalized sector states $\rho_q^{(n)}$ themselves;
3. recovery of the full joint-state information, including cavity coherences.

Only the first goal is generally achievable under the stated constraints.

3 Forward Model and Identifiability

The calibrated analysis sequence is

$$\rho \mapsto U_{\text{wait}}(t) (I \otimes D(\alpha)) \rho (I \otimes D^\dagger(\alpha)) U_{\text{wait}}^\dagger(t), \quad (4)$$

with

$$U_{\text{wait}}(t) = \sum_{m=0}^{N-1} |m\rangle\langle m| \otimes R_z[\phi_m(t)], \quad \phi_m(t) = [\chi m + \chi' m(m-1)] t. \quad (5)$$

For the Fock-diagonal model of Eq. (1), the measured reduced qubit state is

$$\rho_q^{\text{meas}}(\alpha, t) = \sum_{n=0}^{N-1} p_n \sum_{m=0}^{N-1} P_{mn}(\alpha) R_z[\phi_m(t)] \rho_q^{(n)} R_z^\dagger[\phi_m(t)], \quad (6)$$

where

$$P_{mn}(\alpha) = |\langle m|D(\alpha)|n\rangle|^2. \quad (7)$$

Equation (6) implies

$$X(\alpha, t) + iY(\alpha, t) = \sum_{n=0}^{N-1} u_n K_n(\alpha, t), \quad u_n \equiv p_n (X_n + iY_n), \quad (8)$$

with

$$K_n(\alpha, t) = \sum_{m=0}^{N-1} P_{mn}(\alpha) e^{-i\phi_m(t)}, \quad (9)$$

while the longitudinal observable remains

$$Z(\alpha, t) = \sum_{n=0}^{N-1} p_n Z_n. \quad (10)$$

Two consequences are immediate.

- a) **Displacement-only data are uninformative for sector separation.** When $t = 0$, Eq. (6) reduces to the ordinary reduced qubit state because the local cavity unitary disappears under the partial trace.
- b) **The protocol does not separate p_n and Z_n .** Equation (10) collapses all sectorwise longitudinal information into a single scalar.

For the general block model of Eq. (3), the displacement matrix elements mix off-diagonal cavity blocks into the reduced qubit signal:

$$\rho_q^{\text{meas}}(\alpha, t) = \sum_{m,n,r} \langle r|D(\alpha)|m\rangle \langle n|D^\dagger(\alpha)|r\rangle R_z[\phi_r(t)] A_{mn} R_z^\dagger[\phi_r(t)]. \quad (11)$$

Hence cavity coherences are invisible to wait-only data ($\alpha = 0$) but can re-enter once displacement is applied.

4 Numerical Methods

4.1 Single-qubit tomography engine

The reusable single-qubit tomography block begins from the target rotation

$$U = R_x(\pi/3) R_z(\pi/4), \quad (12)$$

acting on $|g\rangle\langle g|$. Three pre-measurement branches are used:

$$U_z = I, \quad U_x = R_y(-\pi/2), \quad U_y = R_x(\pi/2). \quad (13)$$

The reconstructed state is parameterized with a Cholesky factor,

$$T = \begin{pmatrix} e^{s_1} & 0 \\ t_2 + it_3 & e^{s_4} \end{pmatrix}, \quad \rho(T) = \frac{T^\dagger T}{\text{Tr}(T^\dagger T)}, \quad (14)$$

which enforces positivity by construction. We compare least-squares fitting of the measured Pauli expectations to a binomial shot-noise maximum-likelihood objective.

4.2 Protocol grids and inverse solvers

The main inverse study uses four active Fock sectors. The wait-only protocol uses 11 wait times between 0 and $0.35 \mu\text{s}$; the combined protocol uses a 5×9 grid of displacements and waits; the displacement-only protocol uses seven displacements at $t = 0$. For the diagonal model, the recoverable unknown vector is the 9-parameter real vector containing the real and imaginary parts of u_n together with the single scalar $\sum_n p_n Z_n$. Because Eq. (8) is linear in those unknowns, the recoverable fit can be solved by both constrained least squares and binomial maximum likelihood.

4.3 Black-box case library

The study uses three classes of validation cases.

1. **Controlled analytic cases:** ideal block-diagonal SQR outputs, CPSQR-like conditional phases, and coherent companions with the same diagonal sector marginals.
2. **Pulse-level near-diagonal cases:** an optimized multitone black-box waveform synthesized on a six-level cavity truncation with a two-level transmon.
3. **Failure-mode cases:** a short unoptimized waveform and a replay of the optimized waveform on a three-level transmon model to expose leakage.

5 Results

5.1 Single-qubit baseline

Figure 1 shows the fidelity-versus-shot-count scaling of the single-qubit tomography engine. The full-profile study reached a mean maximum-likelihood fidelity of 0.9986 at 10^4 shots per branch, validating the numerical tomography block before it was embedded in the joint inverse problem.

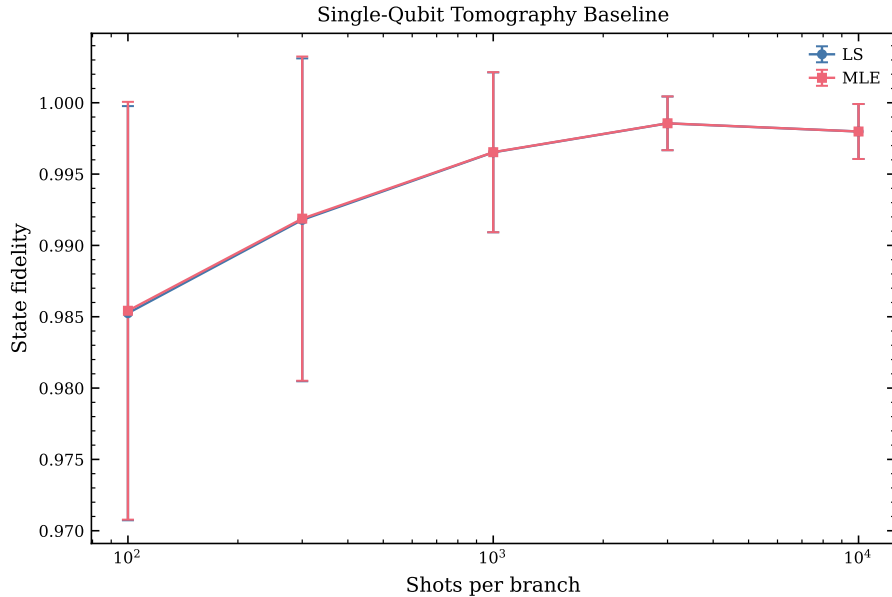


Figure 1: Single-qubit baseline. The Cholesky-parameterized maximum-likelihood reconstruction converges toward unit fidelity with the expected shot-noise scaling and serves as the reusable building block for the Fock-resolved inference layer.

5.2 Protocol identifiability and protocol comparison

The identifiability audit is decisive. Wait-only and combined protocols both have full transverse rank 8 on the recoverable subspace, whereas the displacement-only transverse rank collapses to 2, corresponding to the ordinary reduced-qubit aggregate. In the tested grids the wait-only design is slightly better conditioned (condition number 1.21 versus 2.34), but the combined protocol performs slightly better in noisy finite-shot inversion because it averages over more informative settings.

At 10^4 shots per setting, the mean weighted-transverse root-mean-square error is 4.14×10^{-3} for wait-only, 2.95×10^{-3} for the combined protocol, and 2.17×10^{-1} for displacement-only. Figure 2 summarizes this comparison across the shot sweep.

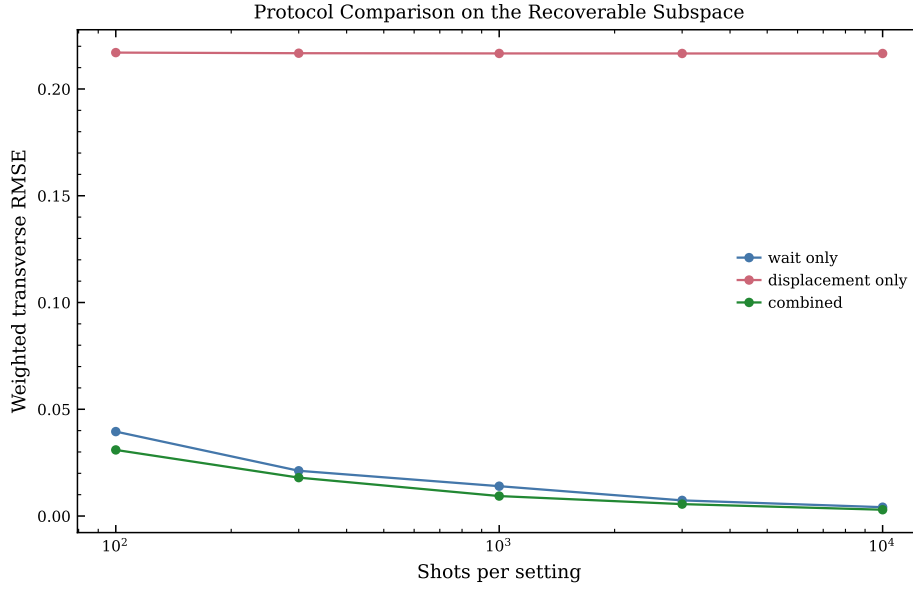


Figure 2: Protocol comparison on the recoverable weighted-transverse subspace. Displacement-only measurements remain non-identifiable at every shot count. Wait-only is already sufficient for recovery, while the combined protocol is modestly more accurate in the tested finite-shot benchmarks.

5.3 Coherence witness and explicit non-identifiability

The strongest positive role of displacement is not the recovery of p_n or Z_n , but the exposure of cavity coherences. In an exact coherent benchmark with the same diagonal sector marginals as the diagonal control case, the wait-only residual remains at 1.6×10^{-16} because the off-diagonal cavity blocks are invisible. The combined residual rises to 1.39×10^{-1} , whereas the diagonal control case remains at 2.8×10^{-16} . The combined protocol therefore functions as a sharp model-mismatch witness.

The negative result is equally strong. For the diagonal benchmark with $z_{\text{tot}} = 0.1162$, we construct two distinct exact gauge-family solutions that reproduce the same recoverable data:

$$\mathbf{p}^{(1)} = (0.4448, 0.2924, 0.1260, 0.1368), \quad \mathbf{p}^{(2)} = (0.3775, 0.3373, 0.1036, 0.1816),$$

even though the original populations are $(0.41, 0.27, 0.19, 0.13)$. Because these distinct sector decompositions yield identical u_n and the same z_{tot} , the protocol cannot claim direct recovery of $\{p_n, Z_n\}$.

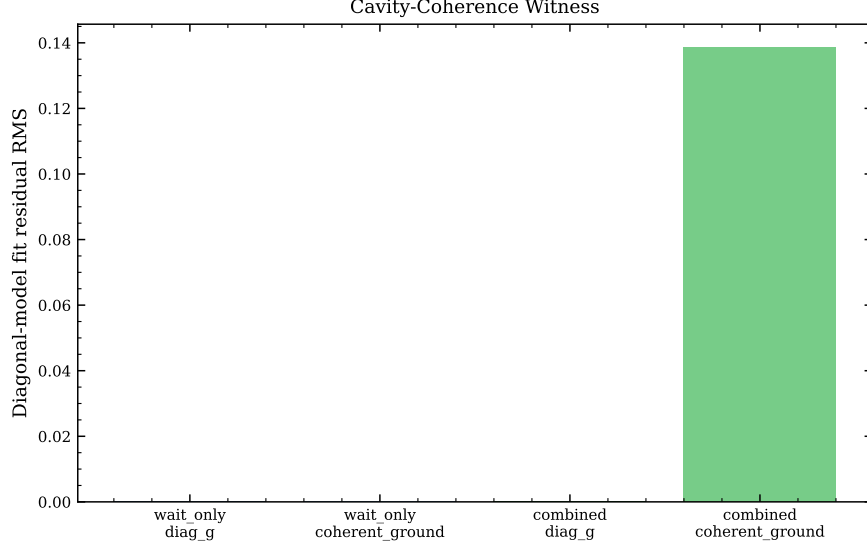


Figure 3: Coherence-witness residuals. Wait-only fits remain exact for both the diagonal and coherent benchmarks because cavity coherences do not enter the reduced qubit data when $\alpha = 0$. The combined protocol remains exact on the diagonal case but produces a large residual on the coherent case, revealing the hidden off-diagonal cavity structure.

5.4 Pulse-level black-box validation

The pulse-level cases reproduce the same qualitative boundary. For the optimized near-diagonal black-box cases, the combined protocol achieves weighted-sector errors below 3×10^{-3} and outperforms wait-only. For example, on the optimized cavity-mixture case with qubit $|g\rangle$, the combined weighted root-mean-square error is 2.60×10^{-3} , compared with 9.63×10^{-3} for wait-only. On the optimized cavity-mixture case with qubit $|+\rangle$, the corresponding errors are 3.38×10^{-3} and 6.70×10^{-3} .

The coherent and leaky cases behave differently for principled reasons. For the optimized cavity-superposition case, wait-only still returns a small weighted error of 8.76×10^{-3} because it is blind to cavity coherence, whereas the combined fit error rises to 1.21×10^{-1} because the diagonal model is no longer valid. For the replay on a three-level transmon, both protocols fail on the recoverable weighted-sector metric, with errors between 2.31×10^{-1} and 2.57×10^{-1} , indicating that leakage moves the output outside the effective qubit-sector model.

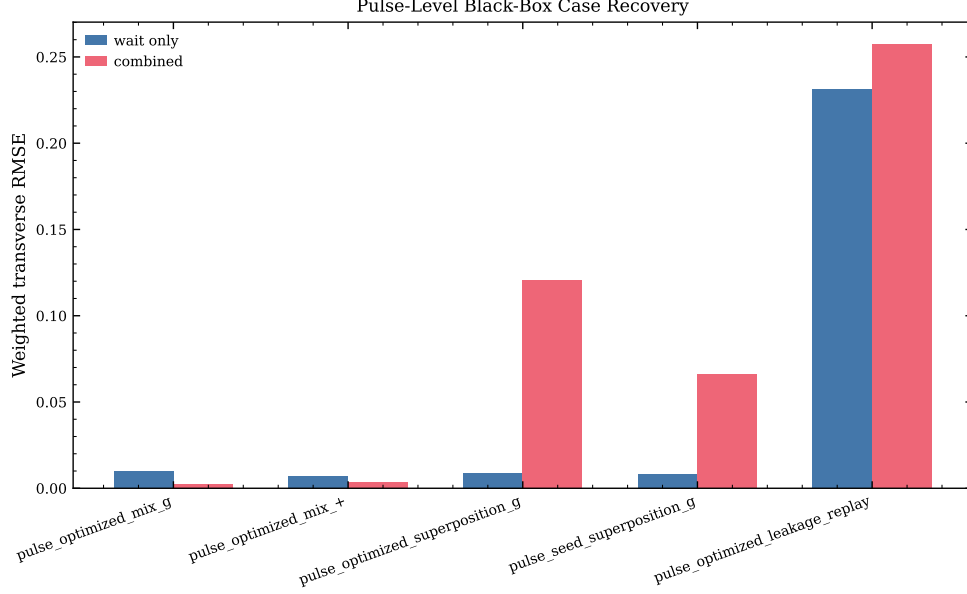


Figure 4: Pulse-level case recovery. Combined inference performs best on the near-diagonal pulse-level black-box outputs, but it deliberately fails on the coherent and leakage-sensitive cases because those outputs violate the diagonal model assumed by the inverse solver.

5.5 Noise robustness and experimental recommendation

Figure 5 summarizes the robustness sweep. Across shot noise, rotation error, displacement calibration error, dispersive mismatch, and wait decoherence, the combined protocol consistently matches or slightly improves on wait-only for the diagonal benchmark. At 1500 shots per setting, the mean weighted-transverse error is 7.52×10^{-3} for the combined protocol and 9.93×10^{-3} for wait-only under shot noise alone; under displacement miscalibration the corresponding errors are 8.13×10^{-3} and 1.17×10^{-2} .

The practical recommendation is therefore two-tiered:

1. Use the combined displacement-plus-wait protocol as the primary experimental workflow when one wants both high-fidelity weighted-transverse sector recovery and an internal witness for cavity coherence or model mismatch.
2. If the combined residual is large, do not interpret the fitted Fock-resolved states literally. In that regime the combined protocol has done its most valuable job by demonstrating that the diagonal effective model has failed. A wait-only fit may still serve as a lower-information diagnostic of the weighted transverse sectors, but not as a full validation of the black-box action.

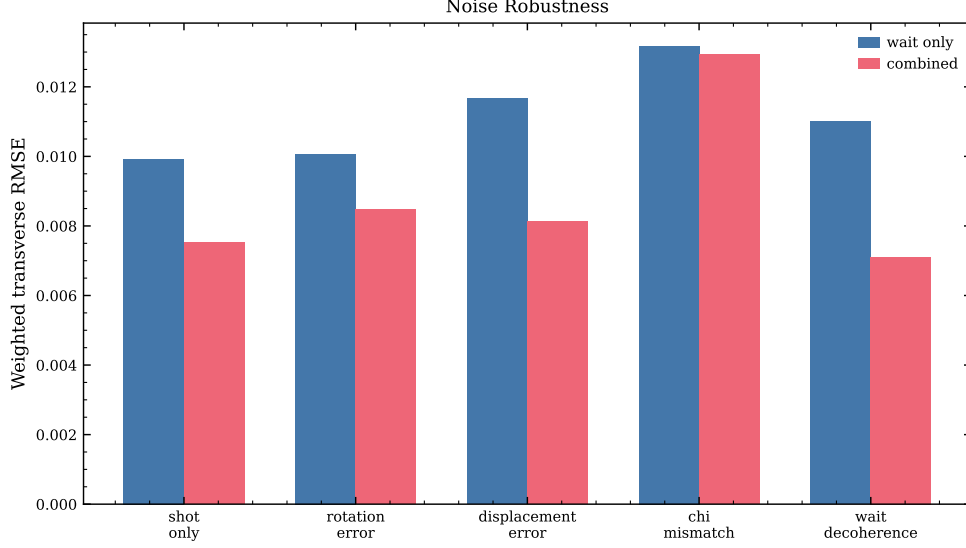


Figure 5: Noise robustness for the canonical diagonal benchmark. The combined protocol is consistently as good as or better than wait-only on the recoverable weighted-transverse metric, but its main additional value is the residual-based coherence witness rather than the recovery of new identifiable parameters.

6 Validation

Three validation gates were enforced.

Sanity checks. The exact wait-only diagonal benchmark fits to machine precision. Displacement-only remains rank-deficient. The coherent benchmark is invisible to wait-only but exposed by the combined protocol.

Convergence. The exact recoverable fit remains at or below 10^{-7} weighted-transverse error under denser combined grids and modest changes in the kernel truncation.

Pulse-level consistency. The near-diagonal pulse-level cases follow the same identifiability logic as the analytic benchmarks, while the leakage replay fails in the expected direction. Because this is an original identifiability and protocol-design study rather than a reproduction of a prior benchmark, a literature-error comparison is not the relevant validation metric here.

7 Limitations and Future Work

The central limitation is structural rather than numerical: the measurement set does not identify p_n or sector-resolved Z_n . No amount of additional least-squares or maximum-likelihood tuning will change that. A future protocol that adds a cavity-sensitive primitive such as parity, selective tagging, or a direct cavity readout could remove that nullspace.

The pulse-level library is also intentionally representative rather than exhaustive. It contains a near-diagonal optimized waveform, a short unoptimized waveform, and a leakage replay. A larger survey over durations, detunings, and decoherence models would sharpen the engineering

recommendation but would not change the identifiability boundary established by the analytic forward model.

8 Conclusion

Black-box SQR validation with calibrated displacement, dispersive waiting, and ordinary qubit tomography is scientifically useful, but only if one states the inverse target correctly. The protocol does *not* recover the full Fock-resolved qubit state in the sense of separate populations p_n and sectorwise longitudinal components Z_n . It *does* recover the weighted transverse sector amplitudes $u_n = p_n(X_n + iY_n)$, and it does so accurately in both analytic and pulse-level near-diagonal cases. Moreover, the combined protocol supplies a valuable residual-based witness for hidden cavity coherence or leakage. The experimentally honest conclusion is therefore that the protocol is trustworthy as a weighted-transverse black-box diagnostic and as a coherence witness, but not as a complete sector-resolved tomography protocol without an additional cavity-sensitive measurement primitive.

A Detailed Results and Data

A.1 Protocol summary

Table 1: Mean weighted-transverse root-mean-square error for the canonical diagonal benchmark under maximum-likelihood fitting.

Shots per setting	Wait-only	Displacement-only	Combined
10^2	3.95×10^{-2}	2.17×10^{-1}	3.10×10^{-2}
3×10^2	2.12×10^{-2}	2.17×10^{-1}	1.80×10^{-2}
10^3	1.40×10^{-2}	2.17×10^{-1}	9.35×10^{-3}
3×10^3	7.33×10^{-3}	2.17×10^{-1}	5.58×10^{-3}
10^4	4.14×10^{-3}	2.17×10^{-1}	2.95×10^{-3}

A.2 Exact gauge-family examples

Table 2: Two exact gauge-family solutions with identical recoverable data for the diagonal benchmark used in the non-identifiability audit.

Solution 1 populations	Solution 2 populations
(0.4448, 0.2924, 0.1260, 0.1368)	(0.3775, 0.3373, 0.1036, 0.1816)

A.3 Representative pulse-level outcomes

Table 3: Pulse-level weighted-transverse errors for representative black-box cases.

Case	Wait-only RMSE	Combined RMSE
Optimized cavity mixture, qubit $ g\rangle$	9.63×10^{-3}	2.60×10^{-3}
Optimized cavity mixture, qubit $ +\rangle$	6.70×10^{-3}	3.38×10^{-3}
Optimized cavity superposition, qubit $ g\rangle$	8.76×10^{-3}	1.21×10^{-1}
Short seed waveform, cavity superposition, qubit $ g\rangle$	7.95×10^{-3}	6.60×10^{-2}
Optimized waveform replay on a three-level transmon	2.31×10^{-1}	2.57×10^{-1}

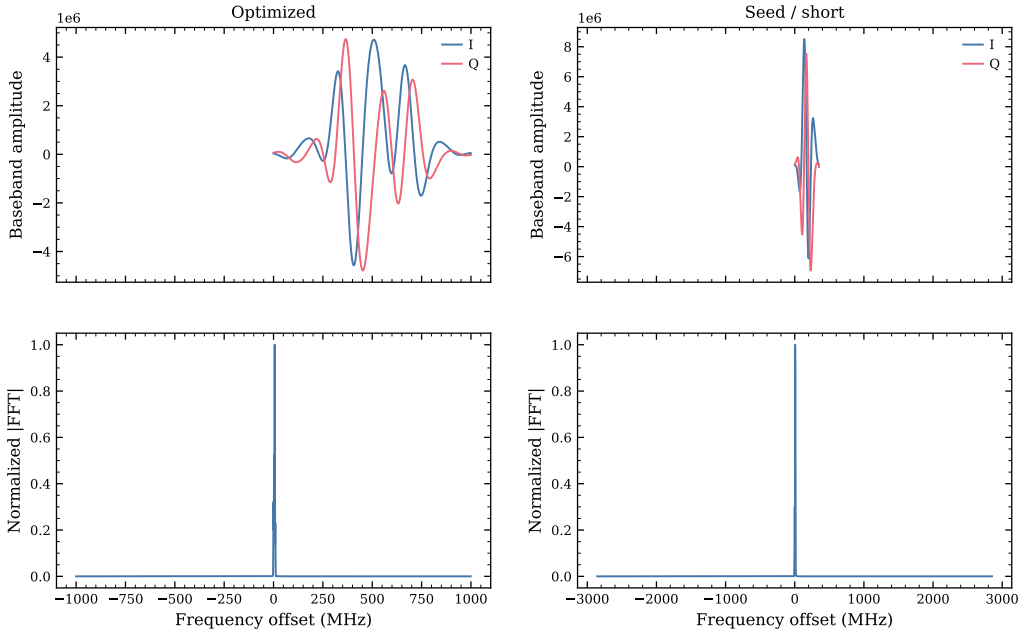


Figure 6: Representative optimized and seed multitone waveforms together with their spectra. These are the pulse-level black-box instances used in the numerical validation study.

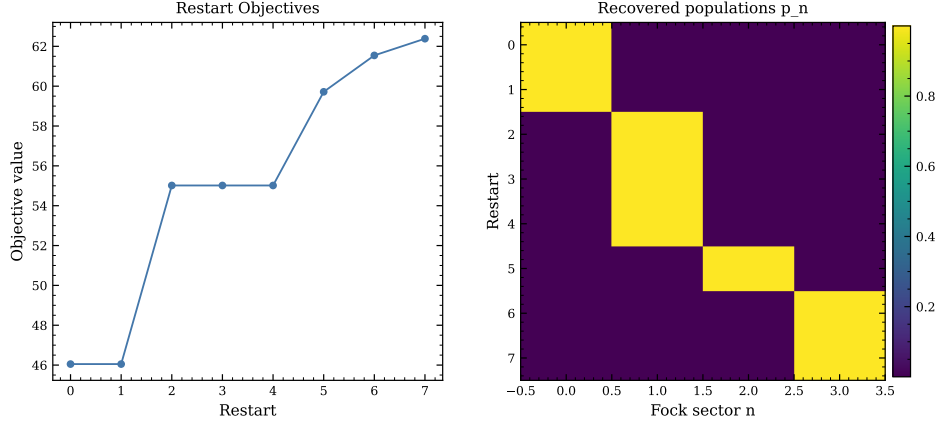


Figure 7: Restart-based full-state fit diagnostic. The broad spread of recovered population assignments complements the explicit gauge-family construction and illustrates the non-identifiability of the full $\{p_n, \rho_q^{(n)}\}$ model under the restricted measurement set.

B Reproducibility

The study is reproducible from the saved machine-readable outputs, figures, and notebook in the study directory. The main execution entry point is `scripts/run_study.py`; the saved outputs are checked by `scripts/validate_results.py`; and the end-to-end interactive workflow is documented in `scripts/reproducibility_notebook.ipynb`. The notebook exposes the main tunable parameters near the top and can rerun the study or simply reload the saved results.

References

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